

Detecting Scams in Job Postings

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1 Introduction and Hypothesis

In today's increasingly digital job market, online platforms like LinkedIn, Indeed, Glassdoor and ZipRecruiter have become essential tools for job seekers. These platforms have made the search for employment more accessible than ever, allowing seekers to discover and filter job postings with greater ease. However, this shift has also introduced new risks -most notably, the rise of fraudulent job postings. These scams have consequences that go beyond simply wasting the job seeker's time, which is especially costly in today's uncertain economy. Fake listings can lead to identity theft, financial scams, malware or phishing attacks, reputational damage for legitimate companies and seekers, emotional distress and missed real opportunities.

Despite efforts by platforms and third-party services to detect and remove these listings, these malicious hiring schemes are becoming increasingly more sophisticated. Additionally, while research on spam and fraud detection has grown in domains like email and financial transactions, employment scams remain relatively underexplored. With this phenomenon in mind, our project aims to develop a model that can detect fake job postings amongst legitimate ones, using a combination of structured (company logo, requirements, salary range) and unstructured features (job descriptions, benefits, company profiles). The dataset we use is the Employment Scam Aegean Dataset (EMSCAD) created by The University of the Aegean.

It's hypothesized that fraudulent job postings will have measurable differences when compared to real postings, and these differences can be leveraged to build a high-performing classifier. In particular, we hypothesize that (a) Text-feature extraction and missingness patterns will serve as powerful feature selection tools; and (b) Lasso logistic regression model will outperform the baseline model in detecting fake job postings.

2 Methods

In this project, we utilize the Employment Scam Aegean Dataset (EMSCAD), which includes real and fake job postings, to build a powerful classification model to detect fraudulent job postings. The dataset contains 18,000 total job postings or descriptions with 800 of them being fake; other relevant information such as job location, requirements, benefits, salary range, etc. Due to missingness in the data and the stark class imbalance, a structured approach to detect fraudulent job postings included a combination of data preprocessing and feature engineering. The methodology spanned three key components: missing data analysis, text feature extraction and model construction. Finally, cross validation was performed to facilitate model’s performance comparison between trivial base-line model and the L1-penalized logistic regression model.

2.1 Missing Data as Feature Engineering

To examine whether missingness patterns could contribute to the classification of fraudulent job postings, missing data across key variables was first quantified. Two engineered columns, “low” and “high” (representing salary range bounds), were excluded from this analysis. Several variables exhibited high missing rates: salary range (84%), department (65%), benefits (40%), required experience (39%), required education (45%), and function (36%). Additionally, binary indicators were created to flag whether each of these variables was missing for a given observation.

The figure below compares the proportion of missing values across key fields between real and fake listings. Fake postings more frequently omit critical details —particularly company profile, required experience, and required education— suggesting a pattern of lower transparency. In contrast, salary ranges are missing more often in real postings, likely due to confidentiality norms.

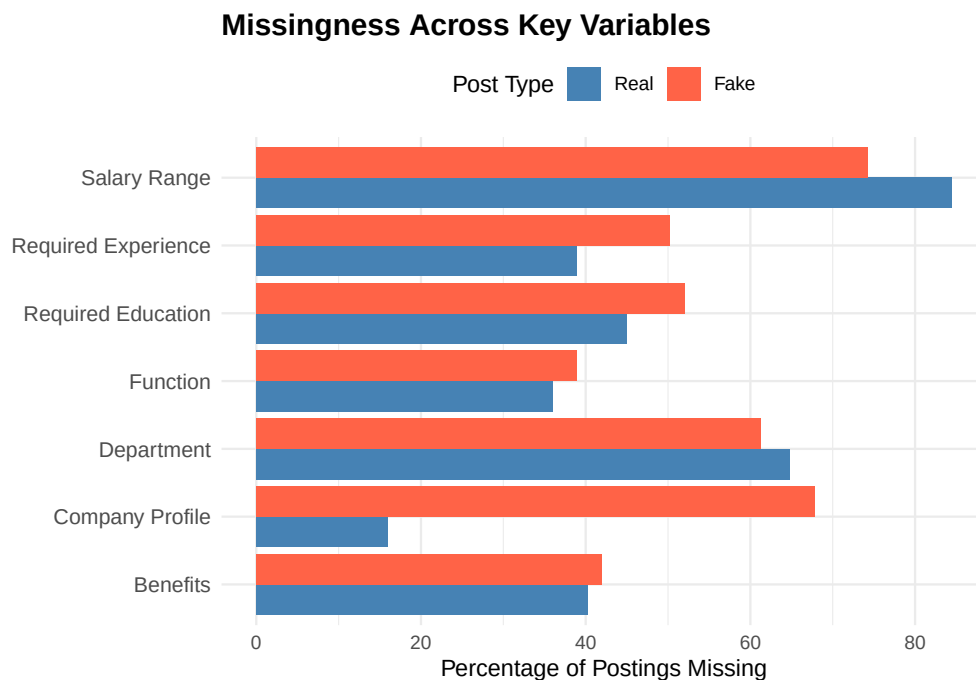


Figure 1: Comparison of missingness percentages across key variables between real and fake job postings.

To assess the significance and effect size of these differences, Chi-square tests and Cramér’s V were calculated. Table 1 shows that the strongest association was observed in company profile ($V = 0.29$, $p < 5.62e-320$), indicating a moderate-to-strong relationship with fraudulent postings. Smaller but statistically significant effects were found for salary range ($V = 0.06$), required experience ($V = 0.05$), and required education ($V = 0.03$). Other variables, such as benefits, department, and function, showed weak or negligible associations ($V < 0.01$). Therefore, the missingness in salary range, required education and required experience were constructed as binary missingness indicators and were included as features in training the classification models.

Table 1: Chi-square test p-values and Cramér’s V for association between missingness and post type (real vs. fake).

Variable	p-value	Cramér’s V
Salary Range	2.11e-15	0.060
Department	4.09e-02	0.016
Benefits	3.42e-01	0.007
Required Experience	3.31e-11	0.050
Required Education	5.02e-05	0.031
Function	8.36e-02	0.013
Company Profile	5.62e-320	0.286

2.2 Text-Based Exploratory Data Analysis (EDA)

Textual content was analyzed to evaluate whether stylistic and linguistic patterns could distinguish fraudulent job postings from legitimate ones. Exploratory analysis was conducted across four text fields: job description, company profile, requirements, and benefits.

2.2.1 Word Count

Total word count comparisons were conducted across four text fields: job description, company profile, requirements, and benefits. To evaluate whether word count distributions differed between real and fake job postings, Wilcoxon rank-sum tests were performed for each field.

Word count features from the description and requirements fields showed statistically significant differences between real and fake job postings (adjusted $p < 0.001$), while word counts for benefits and company profile were not significantly different (adjusted $p = 0.38$ and 0.096 , respectively) as shown in Table 2. As a result, only the two significant fields were retained as predictors for the models.

Table 2: Wilcoxon rank-sum test results comparing word counts by field between real and fake job postings.

Field	W Statistic	p-value	Adjusted p-value	Significance
Benefits	2616902	3.78e-01	3.78e-01	ns
Company Profile	2119154	7.18e-02	9.57e-02	ns
Description	8308902	2.06e-10	4.12e-10	****
Requirements	6415571	4.25e-28	1.70e-27	****

2.2.2 Readability & Sentiment Analysis

To assess whether vocabulary patterns were informative, the most frequent non-stop words were compared between real and fake posts. Despite slight variation, no consistent or meaningful lexical differences were observed at the surface level. Sentiment scores based on positive and negative word ratios were also calculated but showed minimal predictive power ($AUC \approx 0.45$) and were therefore excluded from further analysis. Writing complexity was evaluated using two readability metrics. When modeled with logistic regression, these features yielded an AUC of approximately 0.56, indicating only a marginal relationship between text complexity and likelihood of fraud. As

a result, readability scores were not included in the feature set.

2.2.3 Term Frequency–Inverse Document Frequency (TF-IDF) Modeling

Sparse term-frequency inverse document frequency vectors were generated for each field and evaluated using logistic regression. Results demonstrated in Table 3 suggests strong discriminatory power, with cross-validated AUCs of 0.90 for company profile, 0.82 for description, 0.81 for requirements, and 0.80 for benefits. These findings suggest that while individual words may not differ substantially in isolation, broader textual patterns and rare n-grams carry substantial signal. As a result, TF-IDF representations from all four fields were retained as part of the final model.

Table 3: Mean AUC of TF-IDF across different text fields.

	Field	Mean.AUC
description	description	0.8260
company_profile	company_profile	0.9021
requirements	requirements	0.8182
benefits	benefits	0.8087

2.2.4 Bivariate Analysis of Categorical Variables

To evaluate the relationship between individual categorical features and the likelihood of a job posting being fraudulent, a series of bivariate analyses were conducted. Each categorical variable was cross-tabulated with the fraud label, and Chi-square tests of independence were performed to determine whether the observed distributions were statistically associated with posting authenticity. Cramér’s V was used to assess the strength of these associations.

Several variables showed meaningful associations with fraud status. For instance, the absence of a company logo and certain levels of required education (such as “Some High School Coursework”) were disproportionately represented among fake postings. While some variables, such as location, exhibited visual imbalances in certain regions or country codes (e.g., higher fraud rates in “BH” or “MY”), these were not formally tested and were therefore interpreted descriptively rather than inferentially. Variables that demonstrated either a moderate-to-strong effect size or were conceptually relevant were included as features of the models through one-hot encoding or weight-of-evidence transformation. These included fields such as company logo, required education, and a subset of the most informative location prefixes.

2.3 Results

2.3.1 Model Construction

To build the classification models, a composite feature matrix was assembled by combining three types of predictors: dense numeric features (e.g., the log-transformed salary midpoint and missingness indicators), one-hot encoded categorical variables (such as company_logo and required_education), and sparse TF-IDF vectors derived from four text fields.

As a baseline, a trivial classifier that always predicted the majority class (real) was constructed to illustrate the impact of class imbalance. Two primary models were trained on the resulting feature set: a Lasso-regularized logistic regression and a gradient-boosted decision tree model using the XGBoost framework. XGBoost operates by iteratively building an ensemble of shallow decision trees, where each new tree is fitted to the residuals of the previous ones. Then, cross-validation was used to evaluate model performance on out-of-sample data.

2.3.2 Model Comparison & Visualization

As hypothesized, both the Lasso regression and XGBoost classifiers outperformed the baseline model, which predicted all jobs as real and achieved an AUC of 0.50, indicating no discriminative power. As shown in Table 4 below, the Lasso logistic regression model achieved a strong test AUC of 0.94, with extremely high sensitivity of 99.6%, meaning it correctly identified nearly all real job postings. It also demonstrated high precision for real (96.4%) and fake (91.7%) classifications, reflecting a conservative model that rarely mislabels real jobs. However, its specificity was only 55.0%, indicating it failed to detect nearly half of the fake postings.

Table 4 states that XGBoost achieved a slightly higher test AUC of 0.96, suggesting better overall ranking performance. More importantly, it offered a substantial improvement in fraud detection, with 80.0% specificity, correctly identifying 4 out of 5 fake postings. Its sensitivity remained high at 97.9%, and it achieved even higher precision for real jobs (98.4%). However, its precision for fake predictions dropped to 76.2%, reflecting a more aggressive fraud detection strategy that produced more false positives.

Table 4: Comparison of classification performance metrics between Lasso and XGBoost models.

Metric	Lasso Regression	XGBoost
Test AUC	0.9419	0.9570
Accuracy	0.9621	0.9659
Sensitivity (Recall for real)	0.9959	0.9795
Specificity (Recall for fake)	0.5500	0.8000
Balanced Accuracy	0.7730	0.8898
Kappa	0.6687	0.7620
Positive Predictive Value (Precision for real)	0.9643	0.9835
Negative Predictive Value (Precision for fake)	0.9167	0.7619

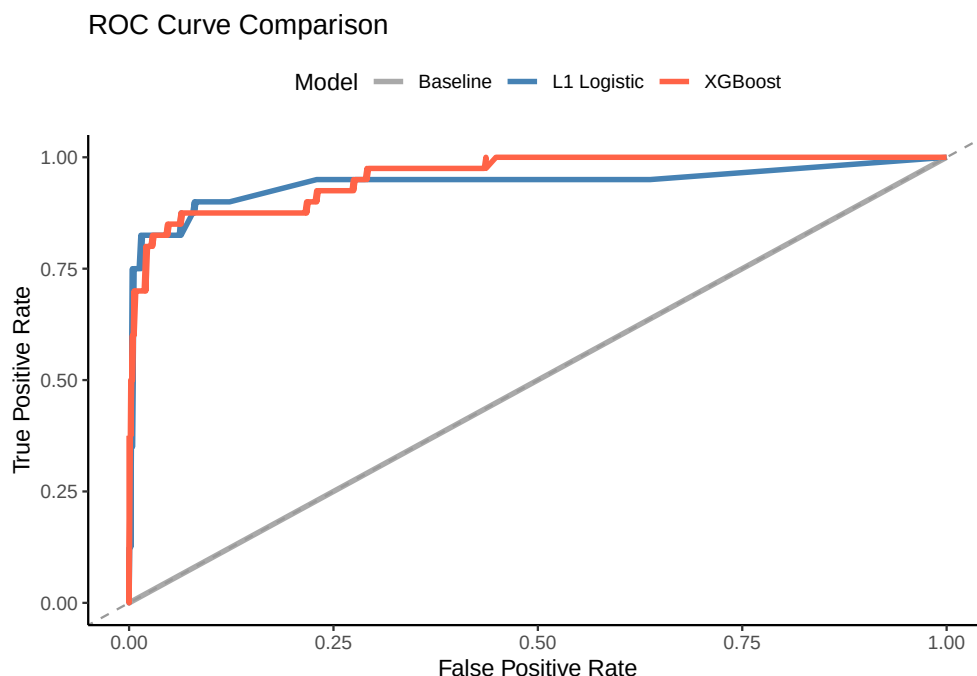


Figure 2: ROC curve comparison between baseline, L1-penalized logistic regression, and XGBoost classifiers.

In figure 2, ROC curve provides a visual summary of these trade-offs. Both Lasso and XGBoost closely follow the upper-left corner of the ROC space, well above the 45-degree baseline. The similarity of their curves confirms that both models perform well overall. However, XGBoost curves slightly higher in the low false-positive region, capturing more true fraud cases earlier. This pattern supports the idea that while both models are strong classifiers, they reflect different trade-offs. Ultimately, model selection should depend on operational priorities. If minimizing false accusations is critical, Lasso is preferable due to its near-perfect sensitivity and high precision. If the priority is detecting as many fraudulent postings as possible, XGBoost is the stronger choice, with higher specificity and balanced accuracy between the two models (88.9% vs. 77.3%) [See Table 4]. These results validate the original hypothesis that engineered features—particularly those derived from text and missingness—enable highly effective fraud detection, with the optimal model

choice shaped by practical constraints.

2.4 Discussion

This project demonstrates that meaningful differences between real and fraudulent job postings exist in this dataset and can be leveraged to train highly effective classification models. Both Lasso and XGBoost achieved strong performance, though each exhibited different strengths. While Lasso was more conservative—minimizing false positives—XGBoost was more effective in surfacing fraudulent cases. These trade-offs highlight an important consideration that model selection should align with platform priorities, whether that’s protecting legitimate users from false flags or maximizing early fraud detection.

Beyond performance, this work shows that missingness patterns and textual structure are valuable sources of signal, even in the absence of user or network-level metadata. However, limitations remain in that the dataset is highly imbalanced, and the models proposed have not yet been tested in live or cross-platform settings. Moreover, neither temporal features nor user behavior data were available, both of which are central to real-world fraud detection systems. Future work should evaluate model generalizability on newer or platform-specific data, integrate additional metadata, and explore hybrid or ensemble approaches that balance interpretability and sensitivity. Incorporating cost-sensitive learning or explainability tools could also help align model behavior with practical deployment constraints.